

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1553

Roanoke Rapids 138 kV
August 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1553. The Interconnection Customer proposes a 148.7 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on August 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 579 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,921,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Big Hanaford Substation - Plaza 138.0 kV	Mitchell Dam - Skaggs Island Jnction 2 345.0 kV	135.5	459	5.4
Big Hanaford - Fritch 138.0 kV	Mitchell Dam - Skaggs Island Jct 2 345.0 kV	135.3	529	9.1
Natchez - Chancellor Row 138.0 kV	Mitchell Dam - Skaggs Island Junction 2 345.0 kV	120.4	367	6.7
Lakeside - Natchez 138.0 kV Line	Skaggs Island Junction 2 - Mitchell Dam 345.0 kV Line	118.4	390	6.7
Redbed Plains Wind Farm - Westcoast Cellufibre 138.0 kV	Mitchell Dam - Skaggs Island Junction 2 345.0 kV L	116.4	380	6.2
Fyffe 345/138 kV TX	Yarnen Substation - Mccartney 345.0 kV	115.4	949	18.4
Lakeside - Fritch 138.0 kV	Mitchell Dam - Skaggs Island Jct 2 345.0 kV Line	113.3	466	6.7
No Mans Land - Yadkin 138.0 kV	Mickens Substation - Mira Vista 345.0 kV	112.7	920	6.2
Yarnen - Mccartney 345.0 kV	Fyffe 345/138 kV TX	111.9	1,656	6.7
Haggin 345-138 kV Transformer	Haggin - North Tower Tap 345.0 kV	109.2	335	5.3
Mira Vista - Mickens 345.0 kV	Toano 138/1kV Autotransformer	108.9	1,472	7.6
Dist 1000 - Birchwood 138.0 kV	Bircwood - Leavitt 345.0 kV	104.6	515	5.1
North Anna - Louisa Generation Facility 345.0 kV	Louisa Generation Facility - Carrabelle 345.0 kV L	104.6	2,511	7.3
Dist 1000 - Trenton Diesel 138.0 kV	Brchwood - Leavitt 345.0 kV L	104.4	527	5.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mira Vista Substation - Yarnen 345.0 kV Line	Toano 138-1 kV TX	101.0	1,859	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Hanaford Substation - Plaza 138.0 kV	Rebuild line with higher-capacity conductor	\$5,447,000	2% (\$102,000)
Big Hanaford - Fritch 138.0 kV	Rebuild line with higher-capacity conductor	\$4,591,000	2% (\$87,000)
Chancellor Row - Natchez 138.0 kV	Rebuild line with higher-capacity conductor	\$1,256,000	2% (\$27,000)
Lakeside - Natchez 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,621,000	2% (\$57,000)
Redbed Plains Wind Farm - Westcoast Cellufibre 138.0 kV	Rebuild line with higher-capacity conductor	\$3,972,000	2% (\$79,000)
Fyffe 345/138 kV TX	Replace transformer with higher-rated unit	\$12,173,000	4% (\$549,000)
Lakeside - Fritch 138.0 kV	Rebuild line with higher-capacity conductor	\$3,861,000	2% (\$84,000)
No Mans Land - Yadkin 138.0 kV	Rebuild line with higher-capacity conductor	\$1,870,000	2% (\$29,000)
Yarnen - Mccartney 345.0 kV	Rebuild line with higher-capacity conductor	\$110,065,000	1% (\$1,003,000)
Haggin 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$14,369,000	3% (\$367,000)
Mickens - Mira Vista 345.0 kV	Reconductor line	\$11,667,000	1% (\$153,000)
Dist 1000 - Birchwood 138.0 kV	Reconductor line	\$8,200,000	3% (\$226,000)
North Anna - Louisa Generation Facility 345.0 kV	Reconductor line	\$32,149,000	1% (\$423,000)
Dist 1000 - Trenton Diesel 138.0 kV	Reconductor line	\$18,074,000	3% (\$499,000)
Mira Vista - Yarnen 345.0 kV	Reconductor line	\$19,699,000	1% (\$236,000)
	Total allocated to project		\$3,921,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.